Movement and kinematics

Introduction

Kinematics is the part of physics that studies **how objects move**, without worrying about why they move.

Example: when you ride a bicycle, you can pay attention to **where you are, what path you follow, how fast you go or if you are accelerating**. All of that is studied by kinematics.

Motion is relative

What does this mean? A person sitting next to you on the bus is at rest from your point of view. But from the point of view of a person sitting at the bus stop, they will see that person moving.

> ❗ Very important
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> Motion always depends on the observer: something moves when it changes its position with respect to a point that we choose (in this case with respect to the observer). This point is the **origin of the reference system** and is part of the **reference point**.

Figure 1: Bus stop

Reference system and position

To understand and describe motion we need some fundamental elements.

Imagine you are talking to a friend so they can find you in a park and you tell them you are one hundred meters from the entrance. With that information you have given a point that acts as the origin of the reference system ("the entrance") and the position ("one hundred meters from that entrance").

Figure 2: 1D reference system

! Very important

- **Reference system**: set of axes and a *point*, which is the **origin**, from which we measure positions.
  - In each problem we will choose a reference system and its origin. In general these will be considered fixed and immobile throughout the problem.
- **Position (x)**: place where an object is with respect to the origin of the reference system at each instant. Unit: **meter (m)**.

Sign convention for position

To describe motion it is important to be clear about the signs of position.

- If we move to the right, the position defined by x will increase towards more positive values.
- If we move to the left, x will take smaller values until zero. If we keep moving to the left, x will take more negative values.

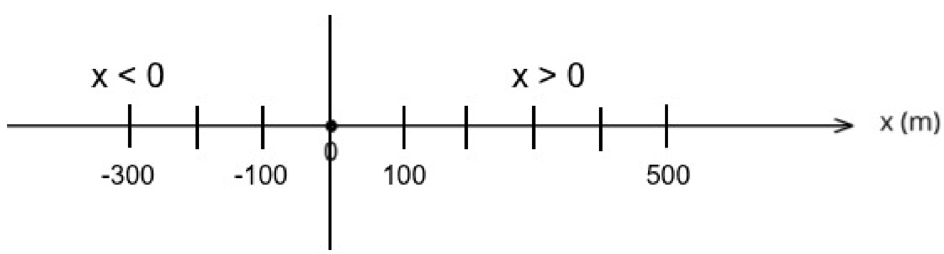

Figure 3: Sign convention for position

What is motion then?

! Very important

We will say that a "body" is in **motion** when it changes its position ($'x'$) with respect to the origin of the reference system over time (t).

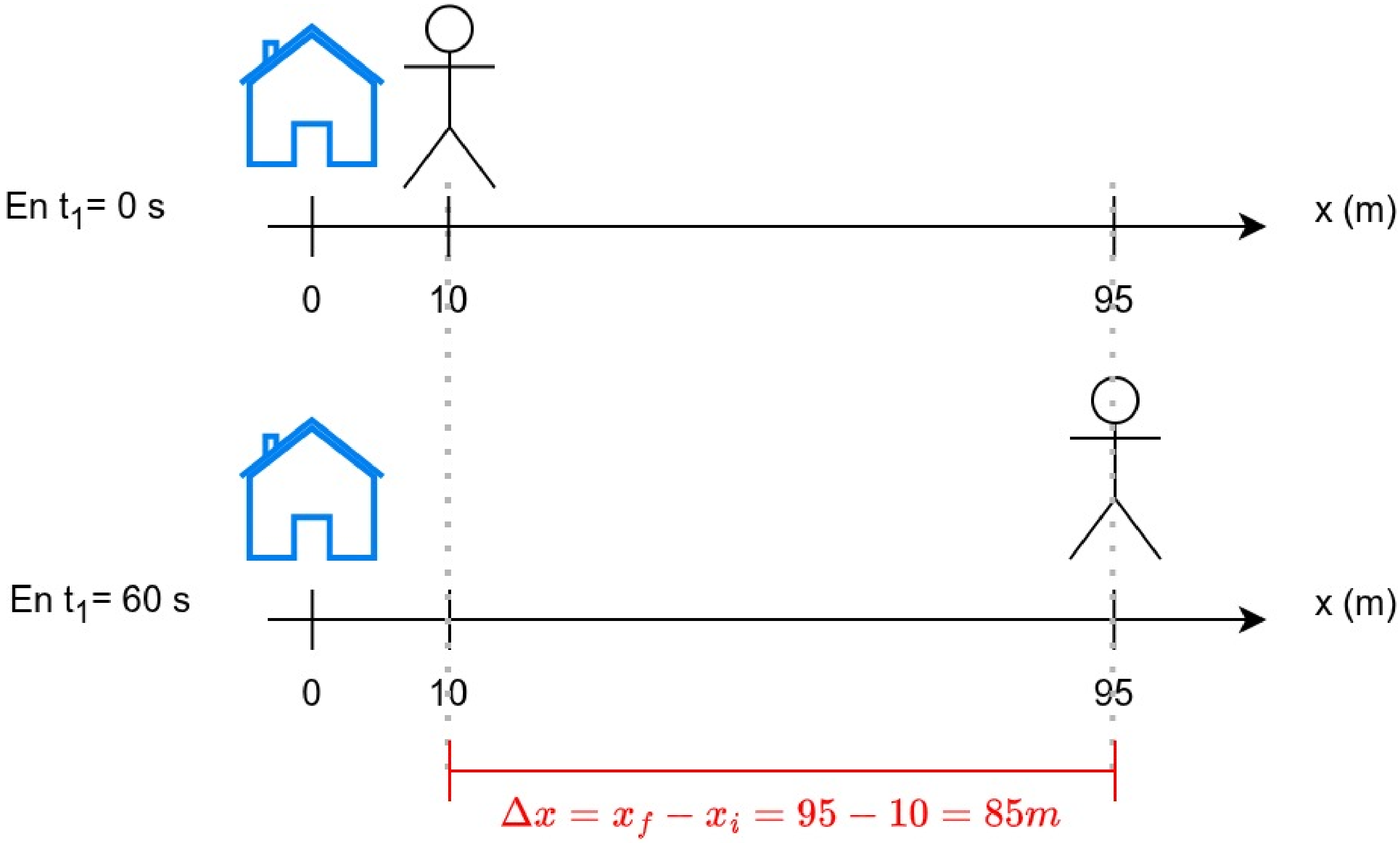

Figure 4: Motion on the horizontal axis with reference system

In the previous image we observe how at time $t1$ the person is at a position $x = 10m$ and after time has passed, that person is at position $x = 95m$.
- They have covered a distance of $d = 95m - 10m = 85m$.
- It coincides with the displacement (which is calculated as final position x_f minus initial position x_i): $\Delta x = x_f - x_i = 95 - 10 = 85m$.
- They did it in the following time: $\Delta t = t_2 - t_1 = 60s - 0s = 60s$.
- Their velocity can be calculated as:

$$v = \frac{\Delta x}{\Delta t} = \frac{85m}{60s} \approx 1,42\frac{m}{s}$$

Trajectory

❗ Very important

Trajectory: is the path followed by an object, that is, the set of its positions as time passes.

For example: the points a cyclist would pass through following the road in the drawing.

Figure 5: Trajectory

Trajectories can be straight (a ball that we drop or a car on a straight road) or curved like the previous image or the shot of a ball into a basket.

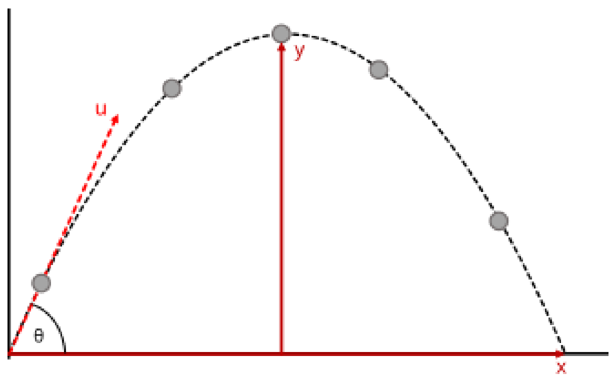

Figure 6: Curved trajectory of a ball when shooting into a basket

> **i** Pay attention!
>
> This course we will only consider straight trajectories. If the trajectory goes through different streets, we will break the final trajectory into different straight segments.

Exercises

1. Define trajectory in your own words.

2. Can the position change without changing the trajectory?

3. Give an example of a curved trajectory.
4. How would you draw the trajectory of a person who wants to go from the School to the Sports Field?

Figure 7: City map

Solutions

1. Path followed by an object.
2. Yes, if it moves within the same type of trajectory.
3. A thrown ball.
4. This exercise has different solutions. The interesting thing is simply to think that they must go through many streets.

Distance traveled and displacement

Very important

Distance traveled (s): is the total distance covered by a body in its motion. Unit: **meter (m)**.
Displacement(Δx): is the distance between the initial position (x_i) and the final position (x_f), it is also measured in **meters (m)** and is calculated as: $\Delta x = x_f - x_i$

When we move in a straight line and do not go back, the distance traveled coincides with the displacement.

But when we go somewhere and come back, these values may not coincide. *For example: sometimes we may have a distance traveled of 100 m, but the displacement is 0. When? When we go from our house to the school which is 50 m away and then return.*

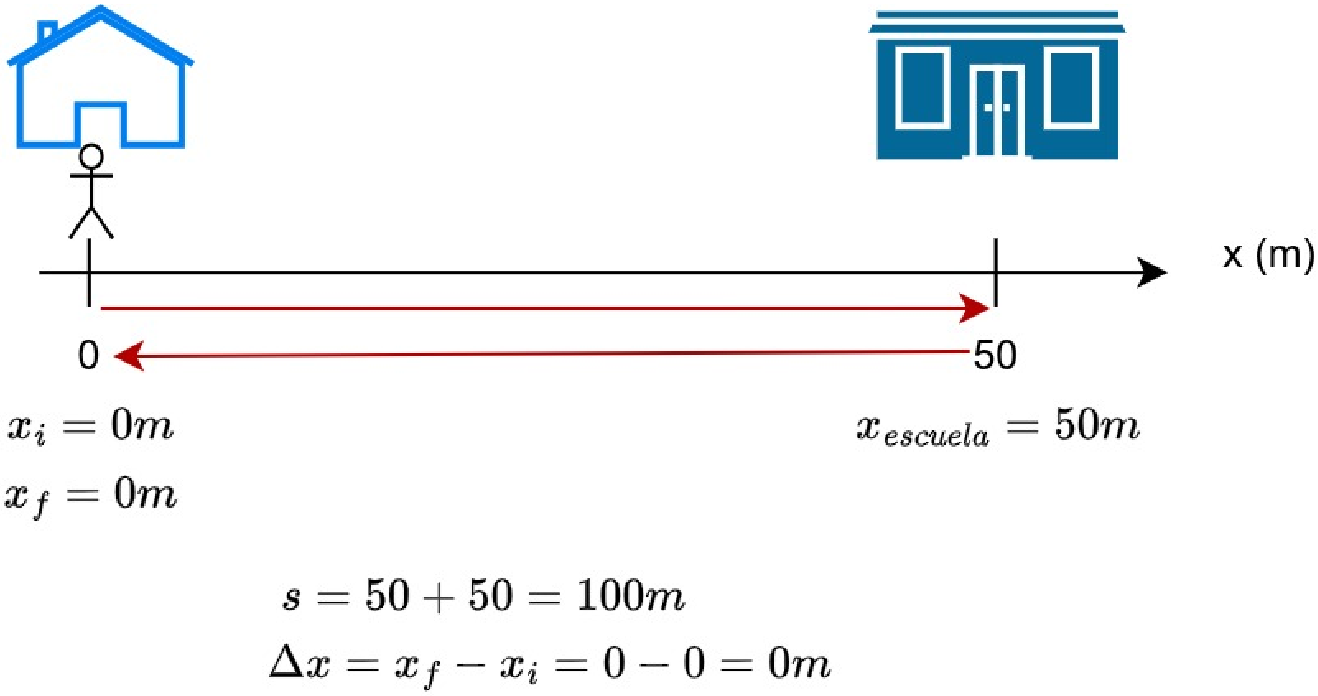

Figure 8: Difference between distance traveled and displacement

Exercises

1. Is distance traveled the same as displacement?

2. Calculate the distance traveled: you go from your house to the school (200 m), from there to the supermarket (another 60 m in the same direction) and from the supermarket back home (260 m from the supermarket to home in the opposite direction).
3. Now calculate the displacement going to the previous places considering that your house is the initial and final point.
4. What would be the displacement if you went from home to the school and from there to the supermarket without returning home?
5. And the distance traveled?
6. Can the distance traveled be zero?

Solutions

1. No.

2. $s = 200 + 60 + 260 = 520m$.
3. $\Delta x = x_f - x_i = 0 - 0 = 0m$.
4. $\Delta x = x_f - x_i = (200 + 60) - 0 = 260m$.
5. $s = 200 + 60 = 260m$.
6. Yes, only when there is no motion, that is, if you do not move.

Additional resources

- English video Motion in one dimension
- Rectilinear motion